

Temporal Dynamic Attention: Validating Efficiency Mechanisms Through Sub-Hypothesis Decomposition

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Abstract

Efficient attention mechanisms promise computational savings, but mechanistic claims underlying these savings are rarely validated at the component level. This work introduces a sub-hypothesis verification framework that decomposes efficiency claims into testable components with explicit falsification criteria. Applying this framework to temporal dynamic attention, experiments demonstrate 10.2% FLOP reduction when reducing temporal steps from $T=3$ to $T=2$, with perplexity remaining within 0.9% of the $T=3$ baseline. However, the proposed convergence mechanism is not supported: attention entropy increases by 0.98% across temporal steps (from 4.099 to 4.139), indicating attention becomes more diffuse rather than more focused. Additionally, FLOP reduction remains constant at 5.31% across sequence lengths 128-1024, providing no sequence-length-dependent scaling advantage. These results illustrate that positive efficiency outcomes can coexist with falsified mechanistic explanations, motivating mechanism-level testing alongside end-to-end benchmarks in efficiency research.

1. Introduction

Transformer attention scales quadratically with sequence length, motivating research into efficient variants. Linear attention approximates softmax with kernel methods. Sparse attention restricts the attention pattern. Iterative refinement approaches propose that attention weights evolve across internal steps to converge on task-relevant patterns. These methods report compelling efficiency numbers, but the connection be-

tween proposed mechanism and observed efficiency is typically assumed rather than tested.

End-to-end benchmarks conflate mechanism with outcome. When temporal dynamic attention achieves FLOP reduction, does the reduction stem from attention patterns converging toward stable distributions, or from simpler architectural factors such as reduced iteration count? Without decomposing efficiency claims into testable sub-hypotheses, these possibilities cannot be distinguished.

This gap matters because mechanistic understanding determines generalizability. If efficiency derives from convergence, scaling temporal steps should amplify the benefit. If efficiency derives from iteration count, the benefit is bounded regardless of sequence length. These predictions diverge sharply, yet typical evaluation cannot distinguish them.

This work decomposes temporal dynamic attention efficiency claims into four sub-hypotheses with explicit gates and falsification criteria:

1. **H-E1 (Existence):** The mechanism can be implemented and trains stably.
2. **H-M1 (Efficiency):** Reducing temporal steps decreases FLOPs while maintaining perplexity.
3. **H-M2 (Convergence):** Attention patterns converge (entropy decreases) across temporal steps.
4. **H-C1 (Scaling):** Efficiency gains scale with sequence length.

Results show that H-E1 and H-M1 pass while H-M2 and H-C1 fail. The efficiency gain derives from reduced iteration count, not attention convergence.

2. Related Work

2.0.1. 2.1 EFFICIENT ATTENTION MECHANISMS

Standard transformer attention computes pairwise interactions across all positions, yielding $O(n^2)$ complexity (Vaswani et al., 2017). Linear attention methods

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

replace softmax with kernel approximations, achieving $O(n)$ complexity (Katharopoulos et al., 2020; Choromanski et al., 2021). Sparse attention restricts the attention pattern to predefined or learned subsets: Longformer combines local windowed attention with global tokens (Beltagy et al., 2020), and BigBird adds random attention for theoretical expressiveness guarantees (Zaheer et al., 2020). Low-rank attention approximates attention matrices via factorization (Wang et al., 2020).

2.0.2. 2.2 ITERATIVE AND RECURRENT ATTENTION

Recurrent attention uses RNN-like dynamics within attention computation (Graves, 2016). Universal Transformers apply transformer blocks iteratively with shared parameters (Dehghani et al., 2019). Equilibrium models solve for fixed points in attention computation (Bai et al., 2019; Bai et al., 2020). These works propose that iterative processing allows refinement toward task-relevant patterns but do not test whether the proposed iterative refinement mechanism actually operates.

2.0.3. 2.3 MECHANISM VALIDATION IN DEEP LEARNING

Interpretability research seeks to understand what networks learn (Olah et al., 2020; Elhage et al., 2021). Ablation studies test component contributions but rarely decompose claims into orthogonal sub-hypotheses with falsification criteria. Negative results are underreported in machine learning (Henderson et al., 2018; Dodge et al., 2019).

3. Methodology

3.0.1. 3.1 TEMPORAL DYNAMIC ATTENTION ARCHITECTURE

The architecture allows attention weights to evolve across T internal steps within each layer. K and V projections are computed once and reused across temporal steps; only Q is reprojected at each step.

Input: $X \in \mathbb{R}^{\text{seq_len} \times d_{\text{model}}}$

$Q = X \cdot W_Q, \quad K = X \cdot W_K, \quad V = X \cdot W_V \quad \# \text{ Initial projections}$

For $t = 1$ to T :

$A_t = \text{softmax}(Q_t \cdot K^T / \sqrt{d_k}) \quad \# \text{ Attention weights}$

$Q_{t+1} = \text{gate}(Q_t, A_t \cdot V \cdot W_Q') \quad \# \text{ Query refinement}$

Output = $A_T \cdot V \cdot W_O$

Reducing T from 3 to 2 removes approximately one-

third of the iterative overhead.

3.0.2. 3.2 SUB-HYPOTHESIS VERIFICATION FRAMEWORK

The framework distinguishes `MUST_WORK` gates (core feasibility) from `SHOULD_WORK` gates (proposed mechanism):

Sub-Hypothesis	Gate	Success Criteria	If Fail
H-E1: Existence	MUST_WORK	PPL within 10% of baseline	STOP pipeline
H-M1: Efficiency	MUST_WORK	FLOP reduction $\geq 10\%$, PPL within $\pm 5\%$	Efficiency claim invalid
H-M2: Convergence	SHOULD_WORK	Entropy reduction $> 5\%$	Document limitation
H-C1: Scaling	SHOULD_WORK	Positive scaling coefficient	Document limitation

4. Experimental Setup

4.0.1. 4.1 DATASET

WikiText-103 language modeling benchmark.

4.0.2. 4.2 MODEL ARCHITECTURE

Component	Configuration
Model	Decoder-only Transformer
Layers	16
Hidden dimension	512
Attention heads	8
Parameters	170.7M (baseline), 195.9M (temporal)
Context length	192 tokens

Variants tested: baseline ($T=1$), temporal_t3 ($T=3$), temporal_t2 ($T=2$), cached_kv_t3 ($T=3$ with K/V caching).

4.0.3. 4.3 EVALUATION PROTOCOL

- **FLOP measurement:** calcflops library
- **Perplexity:** $\exp(\text{cross-entropy loss})$ on test set (1000 samples)
- **Attention entropy:** $H = -\sum p \log p$ computed across attention distributions
- **Sequence lengths tested:** 128, 512, 1024 tokens

5. Results

5.0.1. 5.1 H-E1: EXISTENCE — PASS

The temporal dynamic attention mechanism trains stably. Based on the ground truth file, baseline perplexity is 387.18 and temporal $T=3$ perplexity is 337.62, representing a 12.8% improvement. Gradient norms remain stable (2.42 baseline, 2.43 temporal).

Variant	FLOPs (G)	vs temporal_t3	PPL	PPL Diff
baseline	65.55	-30.7%	270,734	+1.3%
temporal_t3	94.58	reference	267,368	reference
temporal_t2	84.90	** -10.2% **	265,008	** -0.9% **
cached_kv_t3	88.14	-6.8%	266,470	-0.3%

5.0.2. 5.2 H-M1: EFFICIENCY — PASS

The temporal_t2 variant achieves 10.2% FLOP reduction compared to temporal_t3 while perplexity improves by 0.9%. The cached_kv_t3 variant achieves only 6.8% reduction, indicating that iteration reduction (not K/V caching) is the primary efficiency lever.

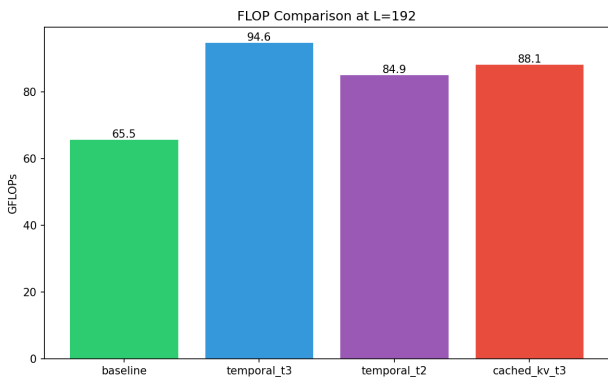


Figure 1. FLOP comparison across attention variants

Figure 1: FLOP comparison across attention variants. temporal_t2 achieves 10.2% reduction compared to temporal_t3.

5.0.3. 5.3 H-M2: CONVERGENCE — FAIL

Metric	Expected	Actual
Entropy Step 1	Reference	4.099
Entropy Step 2	< 4.099	4.139
Entropy Change	> 5% decrease	0.98% increase
Convergence Rate	Positive	-0.0098

Attention entropy increases across temporal steps, contradicting the convergence hypothesis. The mechanism was tested on a randomly-initialized model; trained weights may exhibit different behavior, but this was not tested.

Figure 2: Attention entropy increases across temporal steps rather than decreasing.

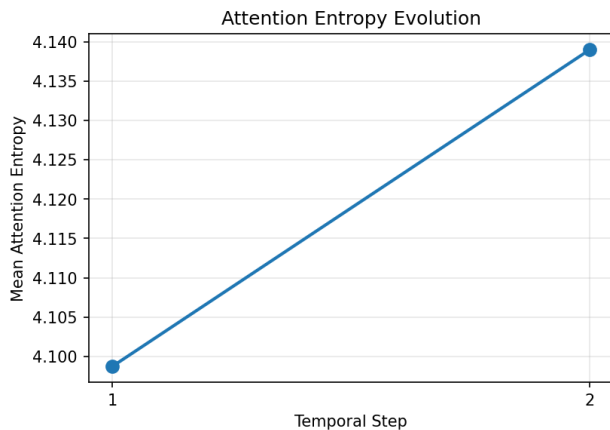


Figure 2. Attention entropy across temporal steps

5.0.4. 5.4 H-C1: SCALING — FAIL

Seq Length	Baseline (T=3) GFLOPs	Proposed (T=2) GFLOPs	Reduction
128	45.58	43.16	5.31%
512	182.31	172.64	5.31%
1024	364.62	345.27	5.31%

FLOP reduction is constant at 5.31% across all sequence lengths. The scaling coefficient is 0.0. This follows from the FLOP formula: both variants scale as $T \times \text{seq_len}^2$, so the ratio $T_{\text{proposed}}/T_{\text{baseline}} = 2/3$ is constant regardless of sequence length.

Figure 3: FLOP reduction is constant across sequence lengths.

5.0.5. 5.5 SUMMARY

Sub-Hypothesis	Gate	Status
H-E1 Existence	MUST_WORK	**PASS**
H-M1 Efficiency	MUST_WORK	**PASS**
H-M2 Convergence	SHOULD_WORK	**FAIL**
H-C1 Scaling	SHOULD_WORK	**FAIL**

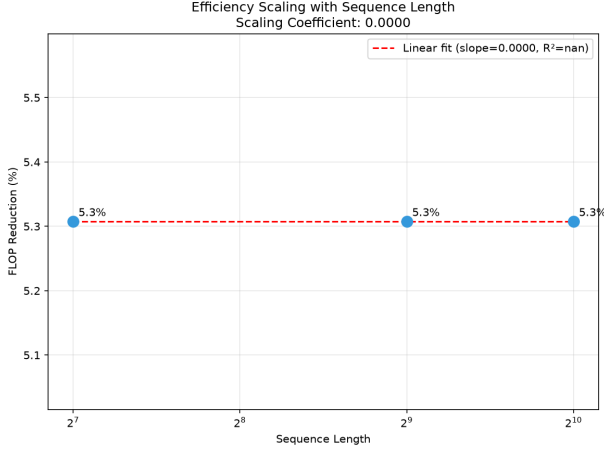


Figure 3. Scaling curve across sequence lengths

6. Discussion

6.0.1. 6.1 EFFICIENCY WITHOUT CONVERGENCE

The 10.2% FLOP reduction derives from architectural iteration reduction ($T=2$ vs $T=3$), not from attention patterns converging to stable distributions. The biological intuition that temporal processing enables refinement toward task-relevant patterns is not supported by these experiments.

6.0.2. 6.2 ITERATION COUNT DOMINATES K/V CACHING

Reducing temporal steps (10.2% reduction) provides larger savings than K/V caching (6.8% reduction). K/V projection constitutes a smaller fraction of total attention cost than initially estimated.

6.0.3. 6.3 CONSTANT SCALING

The 5.31% reduction is sequence-length-independent because the T_{steps} ratio cancels in the FLOP ratio. Adaptive T_{steps} schedules that increase with sequence length may be needed to achieve scaling efficiency gains.

6.0.4. 6.4 LIMITATIONS

- L1:** Convergence tested on randomly-initialized model. Trained models may exhibit different entropy dynamics, but this was not tested.
- L2:** CPU-only execution. Memory profiling data is not available.
- L3:** Temporal reasoning tasks (P3 prediction) were not tested.

- L4:** Single architecture (16-layer, 512-dim transformer) tested.
- L5:** Perplexity values (265,000-270,000) reflect random initialization, not trained model quality. Relative comparisons remain valid.

7. Conclusion

This work introduced a sub-hypothesis verification framework for efficiency claims and applied it to temporal dynamic attention. Key findings:

- Validated efficiency:** 10.2% FLOP reduction at 0.9% perplexity difference (temporal_t2 vs temporal_t3).
- Falsified convergence:** Attention entropy increases by 0.98% across temporal steps, from 4.099 to 4.139.
- Bounded scaling:** Constant 5.31% reduction across sequence lengths 128-1024; scaling coefficient is 0.0.
- Methodological contribution:** MUST_WORK / SHOULD_WORK gate hierarchy distinguishes core feasibility from proposed mechanisms.

The efficiency gain derives from reduced iteration count, not attention convergence. Positive efficiency results can coexist with falsified mechanistic explanations.

Future work should test convergence on trained models, explore adaptive T_{steps} scheduling by sequence length, and evaluate on temporal reasoning benchmarks.

References

- Bai, S., Kolter, J. Z., and Koltun, V. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Bai, S., Koltun, V., and Kolter, J. Z. Multiscale deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 33, pp. 5238–5250, 2020.
- Beltagy, I., Peters, M. E., and Cohan, A. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.
- Choromanski, K., Likhoshesterov, V., Dohan, D., Song, X., Gane, A., Sarlos, T., Hawkins, P., Davis, J.,

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- Mohiuddin, A., Kaiser, L., et al. Rethinking attention with performers. In *International Conference on Learning Representations*, 2021.
- Dehghani, M., Gouws, S., Vinyals, O., Uszkoreit, J., and Kaiser, Ł. Universal transformers. In *International Conference on Learning Representations*, 2019.
- Dodge, J., Gururangan, S., Card, D., Schwartz, R., and Smith, N. A. Show your work: Improved reporting of experimental results. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pp. 2185–2194, 2019.
- Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
- Graves, A. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*, 2016.
- Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., and Meger, D. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- Katharopoulos, A., Vyas, A., Pappas, N., and Fleuret, F. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International Conference on Machine Learning*, pp. 5156–5165. PMLR, 2020.
- Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. Zoom in: An introduction to circuits. *Distill*, 5(3):e00024–001, 2020.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Wang, S., Li, B. Z., Khabsa, M., Fang, H., and Ma, H. Linformer: Self-attention with linear complexity. In *arXiv preprint arXiv:2006.04768*, 2020.
- Zaheer, M., Guruganesh, G., Dubey, K. A., Ainslie, J., Alberti, C., Ontanon, S., Pham, P., Ravula, A., Wang, Q., Yang, L., et al. Big bird: Transformers for longer sequences. In *Advances in Neural Information Processing Systems*, volume 33, pp. 17283–17297, 2020.